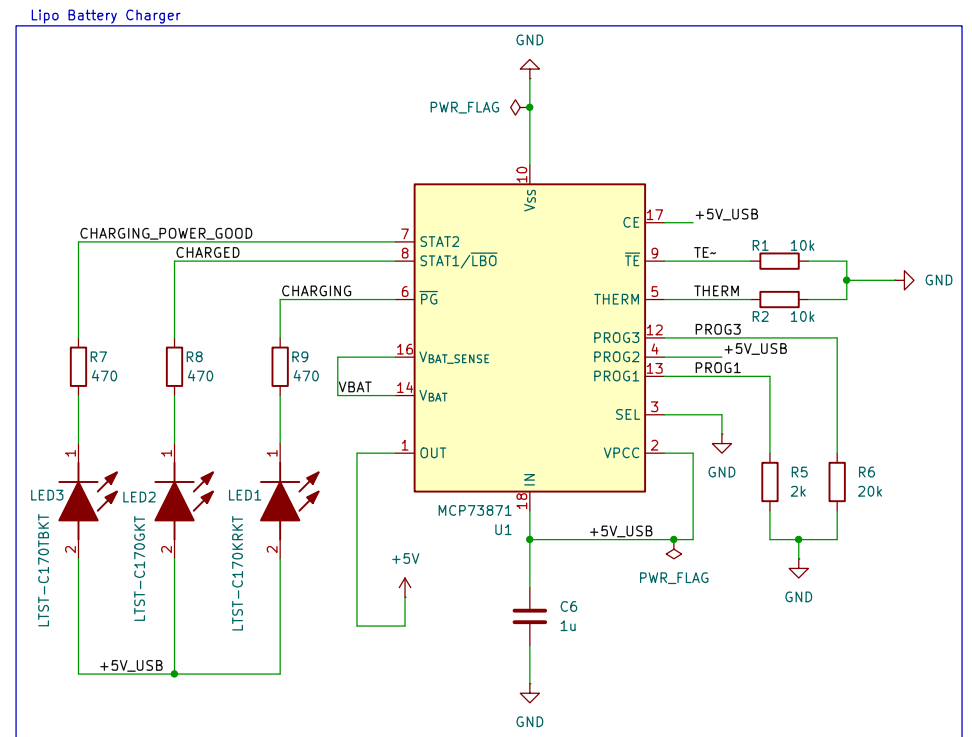
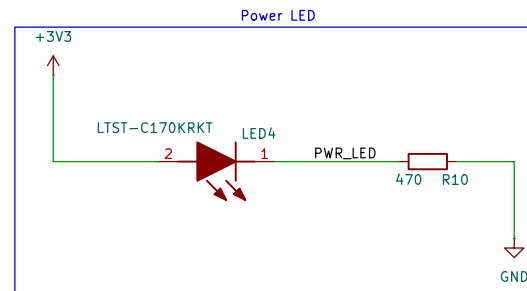
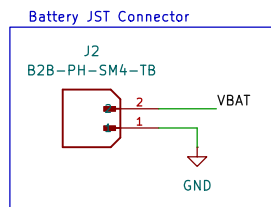
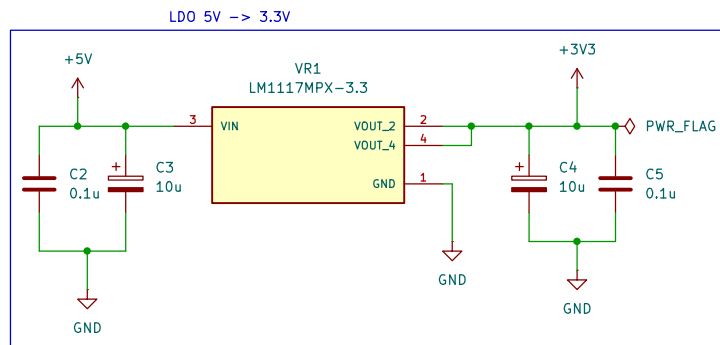
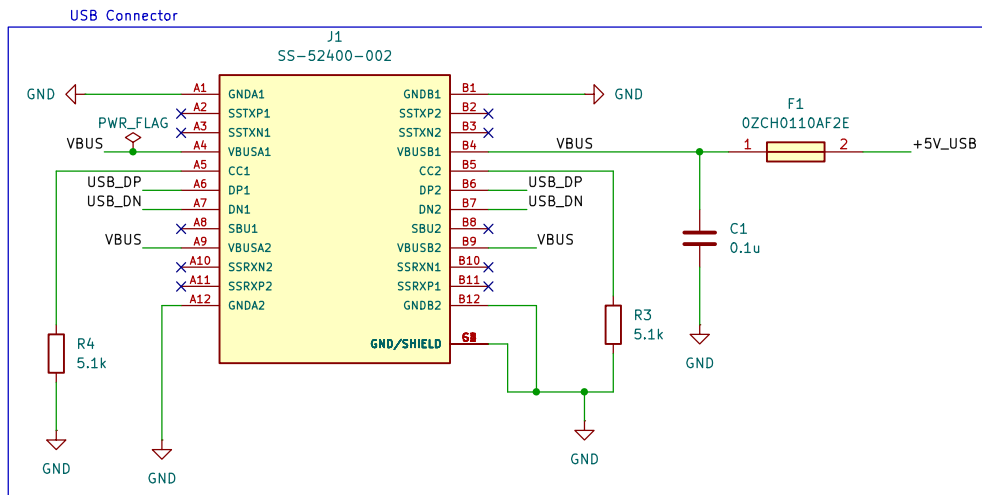
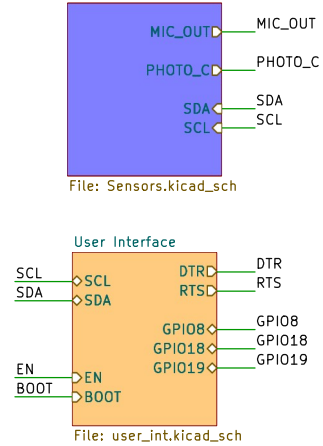


File: ESP32-C3-02.kicad_sch

The diagram illustrates the internal circuitry of a USB-to-serial module. At the center is the CP2102N IC (U4), which interfaces between a USB port and a serial port. The USB side (pins 1-5) is connected to a USB connector (U5) through a USB-DPFD and a USBLC6-25C6 level shifter. The serial side (pins 15-18) is connected to TX_RX and RX_TX pins. The IC also has a WAKEUP pin (13) and a SUSPEND pin (14). The power supply section includes a +3V3 input, a 1k resistor (R11), and a 4.7uF capacitor (C9). The ground connection is labeled GND. The module also features two LEDs (LED5 and LED6) connected to the TX and RX lines through 470 ohm resistors (R14 and R18) and 4.7uF capacitors (C10 and C11). The LEDs are labeled LTST-C170GKT and LTST-C170KRKT. The module is powered by a +3V3 supply, and the ground connection is labeled GND.

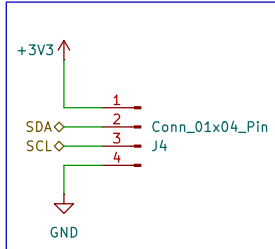


The diagram shows the ESP32-C3-WROOM-02-H4 module (U3) with the following connections:

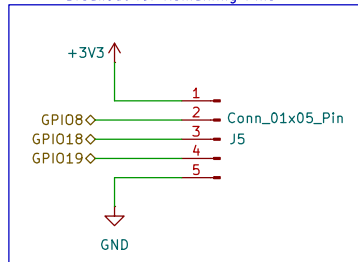
- Power:**
 - EN (2) is connected to a +3V3 supply.
 - IO7 (6) is connected to the +3V3 supply.
 - IO19 (14) is connected to GND.
- SD Card Interface:**
 - Chip select for SD card (blue text) is connected to CS_SD (18).
 - CS- (10) is connected to the chip select line.
 - CS- (13) is connected to the chip select line.
 - CS- (14) is connected to the chip select line.
- Other Connections:**
 - SCLK (R20, 50) is connected to SCL (15).
 - MIC_OUT (4) is connected to SDA (3).
 - RX_TX (12) is connected to TXD (TXD).
 - TX_RX (11) is connected to RXD (RXD).
 - GPIO8 (7) is connected to MOSI-DI (R15, 50).
 - BOOT (8) is connected to MOSI-DI (R15, 50).
 - GPIO18 (13) is connected to MOSI-DI (R15, 50).
 - GPIO19 (14) is connected to MOSI-DI (R15, 50).

The diagram shows a microphone amplifier circuit. A microphone (MK1, CMC-5042PF-AC) is connected to the non-inverting input (IN+) of the MAX4466EXX op-amp (U7). The input signal path includes a coupling capacitor C16 (0.1uF) and a voltage divider consisting of resistors R26 (2k) and R28 (2k) connected to a +3V3 supply. A 1M resistor (R27) is connected from the +3V3 supply to the input node. The op-amp's VCC pin (5) is connected to a +3V3 supply, and its GND pin (2) is connected to ground. The output (OUT, pin 4) is connected to a load (MIC_OUT) through a 10k resistor (R30). A feedback network consisting of a 100k resistor (R31) and a 0.1uF capacitor (C20) is connected from the output back to the inverting input (IN-, pin 3). A 1M resistor (R29) is also connected from the inverting input to ground. A 0.1uF capacitor (C18) is connected from the output to ground.

OLED Display Connector (I2C)



Breakout for Remaining Pins



Reset and BOOT Buttons

